

No need to submit. Please attempt all these questions yourself and reach out for any difficulties. Most of these questions are from [3] and [1] (available here)

- Let $\mathcal{P}_n$ be the set of all polynomials of degree at most n in the variable x , with real coefficients. So, any $p \in \mathcal{P}_n$ is of the form

$$p(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n, \quad \text{for } a_i \in \mathbb{R}.$$

- Verify that $\mathcal{P}_n$ is a vector space over $\mathbb{R}$.
- Are the vectors $\{1, x, x^2, \dots, x^n\}$ linearly independent?
- What is the dimension of $\mathcal{P}_n$?

- Consider the set of numbers of the form

$$\mathcal{S} = \{a + b\sqrt{2} + c\sqrt{3} + d\sqrt{6} : a, b, c, d \in \mathbb{Q}\}.$$

- Is $\mathcal{S}$ a vector space over $\mathbb{Q}$? Can you find a basis for this space?
- is $\mathcal{S}$ a vector space over $\mathbb{R}$?

- Problems 1-6 (Chapter 1) from the book [2], pre-print available at

<https://drive.google.com/file/d/1es9rfHw3xraAdJPg8CJkEljJvwIejk-v/view?usp=sharing>.

- Consider the set of signals $y(t)$ that satisfy the following equations. In which case(s) does this set of signals form a vector space (assuming the standard function addition and scaling operations)?

$$(a) \quad (t^2 - 3)\frac{dy}{dt} + 2ty(t) = 0$$

$$(b) \quad 3\frac{d^2y}{dt^2} + 2\frac{dy}{dt} + 4y(t) = 0$$

$$(c) \quad \left(\frac{dy}{dt}\right)^2 + 2y(t) = \exp(t)$$

- Consider $\mathcal{P}$ as the space of periodic signals $f : \mathbb{R} \mapsto \mathbb{R}$ of period 1, with the inner product

$$(f, g) = \int_0^1 f(x)g(x)dx;$$

as discussed in class. We know that $\mathcal{P}$ is a vector space over $\mathbb{R}$.

- Show that any cosines of different integer frequencies are orthogonal, i.e.

$$\cos(2\pi mx) \text{ and } \cos(2\pi nx) \text{ are orthogonal for any } m \neq n \in \mathbb{Z}$$

- What is the dimension of $\mathcal{P}$?
- Does orthogonality from (a) continue to hold if m, n are made non-integer?
- Consider the sequence of functions $(f_n^k : [0, 1] \mapsto \mathbb{R})$

$$f_n^k(x) = \begin{cases} 2^{n/2} & \text{for } k-1 \leq 2^n x \leq k-0.5 \\ -2^{n/2} & \text{for } k-0.5 \leq 2^n x \leq k \\ 0 & \text{otherwise.} \end{cases}$$

Here $1 \leq k \leq 2^n$ and $k, n \geq 0$ are non-negative integers. Sketch these functions for a few n, k . Note that each of these functions can be thought of as being in $\mathcal{P}$ by simply extending them periodically outside $[0, 1]$.

- Are the functions f_n^k for $1 \leq k \leq 2^n$ orthogonal?
- Intuitively, do you expect $\{f_n^k\}_{k,n=1}^\infty$ to be a basis for $\mathcal{P}$?

References

- [1] Stephen Boyd and Lieven Vandenberghe. *Introduction to applied linear algebra: vectors, matrices, and least squares*. 2018.
- [2] Brad G Osgood. *Lectures on the Fourier Transform and Its Applications*, volume 33. American Mathematical Soc., 2019.
- [3] Ian Stewart and David Tall. *The foundations of mathematics*. OUP Oxford, 2015.